

HelixTranslate — Working Plan: Unfinished Items & Known Issues

Revision: 2 **Last modified:** 2026-06-15T04:45:00Z **Authority:** operator mandate 2026-06-14/15 ("what is left unfinished and what are (known) issues ... nothing left unfinished with zero issues when we finally finish") **Scope:** the complete, honest (§11.4.6) inventory of everything not-done + every known issue, structured as a subagent-driven execution plan (§11.4.70). Each item carries: WHAT, STATUS, EVIDENCE, SUBAGENT-TASK, ACCEPTANCE (rock-solid proof §11.4.123, no bluff), and PARALLEL-SAFE (disjoint scope for §11.4.58/§11.4.119).

Baseline at authoring: main HEAD e4c96f6; go build ./... clean; full go test ./... -p 1 GREEN at quiescent checkpoints; **38 genuine, reproduce-first, mutation-proven bug fixes** landed this session (waves 1–12) across the main module + 8 owned submodules, all committed + pushed (submodules → own mirrors, main → both upstreams, gitlink pointers synced). The high-yield product bug-hunt has reached the §11.4.118 **completeness boundary** — recent waves return clean non-findings on already-hardened/rule-correct packages.

Status legend: OPEN · OPERATOR-BLOCKED (needs credentials/decision) · DESIGN (needs architecture decision) · AUTONOMOUS (a subagent can close it now) · DONE.

P0 — Release blockers (must close before any release tag)

P0.1 — Version single-source-of-truth [DONE – value still operator-gated, see P1.0]

- **WHAT:** pkg/version/app.go now holds const AppVersion = "2.3.0"; the 5 binaries were reconciled to it this session. VERSION=2.3.0 is authoritative per CLAUDE.md; Makefile still references 3.0.0.
- **REMAINING:** the *number* (2.3.0 vs 3.0.0) is P1.0; the Makefile literal should track the chosen value.
- **ACCEPTANCE:** all appVersion literals == pkg/version.AppVersion (DONE); Makefile reconciled once P1.0 decided.

P0.2 — Full §11.4.40 release retest not yet run [OPEN – sequenced last]

- **WHAT:** Release requires the complete §11.4.40 7-step retest (pre-build sweep, post-build sweep, on-device/topology cycle, meta-test mutation sweep, Challenge bank sweep, Issues/Fixed audit, CONTINUATION sync) on a clean

baseline. Repeated full go test ./... -p 1 sweeps are GREEN, but the full 7-step ritual has not been run end-to-end.

- **SUBAGENT-TASK:** sequenced AFTER P1.0 (version) + P4 (gate suite) + P6 (Challenges). Then run all 7 steps with captured evidence.
- **ACCEPTANCE:** all 7 steps GREEN with captured evidence + operator confirmation.

P0.3 — No §11.4.151 prefixed release tag yet [OPERATOR-BLOCKED on P0.2 + P1.0]

- **WHAT:** No release tag exists. Per §11.4.151 it must be helix_translate-<version> (prefix from HELIX_RELEASE_PREFIX env, else lowercased root dir) on main repo + every owned submodule, fanned to all upstreams (no force §11.4.113).
 - **ACCEPTANCE:** prefixed tag created post-P0.2, pushed to all upstreams across main + owned submodules.
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P1 — Operator / credential-gated (cannot be closed autonomously)

P1.0 — Decide the authoritative version number [OPERATOR-BLOCKED]

- **WHAT:** Next version = 2.3.x or 3.0.0? CLAUDE.md says VERSION (2.3.0) wins; Makefile says 3.0.0. **Recommendation (§11.4.101 evidence-backed):** 2.3.0 per the CLAUDE.md authoritative-VERSION rule, unless you intend a major bump.

P1.1 — Provider credentials absent/invalid [OPERATOR-BLOCKED]

- **OPENAI_API_KEY** absent → allowlist stale (missing gpt-4o-mini/4.1/o-series). **ANTHROPIC_API_KEY** absent → allowlist very stale (missing 3.5/3.7/4.x). **GEMINI_API_KEY** invalid (live /models "API Key not found"). **ZHIPU** out of balance (error 1113) → allowlist stale (live glm-4.5...5.1) but translation/response-shape unverifiable.
- **UNBLOCK:** operator adds/refreshes keys / recharges Zhipu → then the proven **deepseek pattern** applies (live /models → verify-translate + verify response-content shape → additive allowlist update + RED-proven gate). Template: deepseek fix in pkg/translator/llm/llm.go.

P1.2 — ~30 other provider allowlists not audited [OPERATOR-BLOCKED on keys]

- **WHAT:** qwen/groq/cohere/mistral/xai/replicate/cerebras/cloudflare/siliconflow/hyperbolic/togetherai/sambanova/kimi/novita/nlpcloud/upstage/sarvam/modal/publicai/nia/vulavula allowlists unverified vs live models.
- **UNBLOCK:** funded keys per provider → deepseek pattern per provider (verify-translate + string-content shape before adding).

P2 — Design decisions (need architecture/operator input; not blind autonomous fixes)

P2.1 — Inert CLI flags [DESIGN]

- `cmd/unified-translator` parses `-chunk-size/-workers/-concurrency/-verify` but never consumes them (chunking is automatic + correct); `-monitoring` start is a print-only stub. DECISION: wire to real behavior OR remove (removal needs §11.4.122 operator confirm). Then RED→GREEN.

P2.2 — Inert config fields [DESIGN]

- `D0CXConfig.MinTextLength/IgnoreStyles` + `PDFConfig.MinTextLength` declared but unconsumed. DECISION: wire (changes output, needs care+tests) or drop.

P2.3 — Verifier `MinScoreThreshold` scale inconsistency [DESIGN]

- `pkg/api` handler compares threshold vs raw 0-100 `OverallScore`; the adapter compares vs normalized 0-10. Contracts contradict. **NOTE:** wave-7 fixed the adapter's `GetPreferences-drops-all` + `empty-cache` bugs, but the cross-layer scale contract itself is still a design call — declare canonical scale (0-100 or 0-10) before wiring either path (fixing one side blindly breaks the other's test, §11.4.120).

P2.4 — Reasoning-model structured-content support [DESIGN]

- OpenAI-compatible clients assume content is a `STRING`; some reasoning models return a structured `LIST` (verified Mistral `magistral-medium-latest`; likely `glm-5/deepseek-reasoner`). Adding structured-content handling unlocks them. Non-trivial; design+tests.

P2.5 — Markdown not a first-class CLI input [DESIGN]

- `.md` input detected as `TXT` (works, structure not preserved). First-class markdown input is an enhancement.

P2.6 — `cmd/translator` intermediate-markdown download-dir inconsistency [OPEN, needs live SSH]

- Intermediate `.md` downloads to `Dir(InputFile)` vs `Dir(OutputFile)` across paths; reproducible only under live SSH. SUBAGENT-TASK: reproduce via §11.4.76 Containers submodule (boot SSH worker container) → RED → fix → GREEN. Gated on containerized SSH infra.
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P3 — Dead / unwired code (§11.4.124 investigate-before-remove)

P3.1 — pkg/hash package [AUTONOMOUS – investigate] · PARALLEL-SAFE

- **WHAT:** pkg/hash has ZERO importers across the module. §11.4.124: must NOT be removed without git-history investigation (where/how wired, when it died, hidden-reference check) + §11.4.122 operator confirm (shipped package).
 - **SUBAGENT-TASK:** `git log --follow/-S pickaxe/blame` → capture FACT → either wire it in properly + add tests, OR (proven-unneeded) remove in its own descriptive commit citing git evidence + operator confirm.
 - **NOTE:** an untracked compiled hash binary + workable-items binary sit at repo root (build artifacts, §11.4.30 — never commit; candidate for .gitignore entry, see P4.4).
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P4 — Governance / constitution-compliance (partially landed this campaign)

P4.1 — Workable-items tracker constellation [PARTIAL – DONE-scaffold, ongoing sync]

- **DONE:** docs/Issues.md (10 open ATM tickets), docs/Fixed.md, docs/Issues_Summary.md, docs/Fixed_Summary.md, docs/CONTINUATION.md, the §11.4.93 Go cmd/workable-items tool + tracked docs/workable_items.db (§11.4.95) all exist (created this campaign).
- **REMAINING:** the 38 campaign fixes are recorded narratively in CONTINUATION rev 52 but NOT all migrated into Fixed.md/the DB with per-fix ATM IDs; procedure docs docs/procedures/issues/*.md (§11.4.63) + per-item Reopens/Status docs not yet present.
- **SUBAGENT-TASK:** migrate the 38 fixes into Fixed.md + workable_items.db with ATM IDs + Status/Type; add the 5 procedure docs. PARALLEL-SAFE (docs-only).

P4.2 — Pre-build CM-* gate suite [PARTIAL – 8 gates landed, more mandated]

- **DONE:** scripts/pre_build_verification.sh with 8 gates (CM-GITIGNORE-PRECOMMIT-AUDIT, CM-NO-FAKES-BEYOND-UNIT, CM-SCRIPT-TARGET-SHELL-PARSEABLE, CM-VERSION-SINGLE-SOURCE, CM-TRACKER-DOCS-PRESENT, CM-ATM-TICKET-IDS, CM-DOC-SIBLING-SYNC, CM-NO-FORCE-PUSH-ABSOLUTE), each with a paired §1.1 mutation, all green.
- **REMAINING:** the constitution references dozens more CM-* gates (regression-guard-registered §11.4.135, multi-pass-evaluation, sink-evidence-per-feature, etc.). **SUBAGENT-TASK:** implement the next highest-value gates each with a paired mutation. PARALLEL-SAFE (scripts-only).

P4.3 — §11.4.65 universal markdown export [AUTONOMOUS – verify] · PARALLEL-SAFE

- The doc-sibling guard (CM-DOC-SIBLING-SYNC) is green; commit wrapper auto-syncs. SUBAGENT-TASK: full audit that every tracked non-source .md (incl. this WORKING_PLAN.md) has fresh .html+.pdf siblings.

P4.4 — .gitignore build-artifact hygiene [AUTONOMOUS] · PARALLEL-SAFE

- **WHAT:** untracked compiled binaries hash, workable-items (and tracked rebuilt unified-translator/preparation-translator/translate-ssh) sit at repo root. §11.4.30 forbids committing build artifacts. SUBAGENT-TASK: add .gitignore entries for the compiled root binaries + a §11.4.77 regen note (they rebuild from cmd/); leave the tracked ones for an operator decision (removing a tracked binary is a visible change). PARALLEL-SAFE.

P4.5 — commit_all.sh companion doc (§11.4.18) [AUTONOMOUS, minor] · PARALLEL-SAFE

- This session patched scripts/commit_all.sh (§11.4.120 guard reconcile). §11.4.18 wants a docs/scripts/commit_all.md companion; none exists. SUBAGENT-TASK: author the companion doc (Purpose/Usage/guard behavior) if the project adopts §11.4.18, else record a deliberate deviation.
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P5 — Owned submodules (§11.4.28 equal-codebase) [DONE this campaign]

P5.1 — Submodule bug-hunt [DONE – 20 mutation-proven fixes, pushed]

- **DONE:** llm_provider ×5, llms_verifier ×4, doc_processor ×2, llm_orchestrator ×2, docs_chain ×2, challenges ×2, containers ×2 — all reproduce-first + mutation-proven, committed + pushed to their own mirrors, parent gitlink pointers synced (commit ebea2d3+e21c779). vision_engine's gocv bug was independently fixed upstream (64b7d08) → integrated, not double-fixed.
- **REMAINING:** helix_qa — NOT hunted; another session is actively working it (m working tree, §11.4.119 single-owner). security + constitution submodules — not hunted (constitution is governance; security submodule is a candidate next pass). SUBAGENT-TASK (after operator confirms no collision): hunt the security submodule.

P5.2 — §11.4.147 misplaced sibling clone cleanup [OPERATOR-BLOCKED]

- **WHAT:** a separate sibling clone /Volumes/T7/Projects/llm_orchestrator (OUTSIDE the parent tree) holds a STALE, superseded, uncommitted RoundRobinSelector fix (M pkg/agent/multi_pool.go + new test) — a

subagent wrote it there by mistake; the correct fix was re-applied + committed in the submodule (4e08b93). The sibling's uncommitted change is now obsolete.

- **EVIDENCE:** `git -C /Volumes/T7/Projects/llm_orchestrator status` shows the stale M/?? entries.
 - **SUBAGENT-TASK:** none — operator decides whether to `git checkout -- . /` discard that sibling clone's working tree (safe: the fix is already in the real submodule). Flagged so it is not mistaken for live work.
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P6 — Full test-type coverage (§11.4.25 / §11.4.27) **[OPEN, large]**

P6.1 — Per-feature test-type matrix + Challenges + HelixQA **[OPEN]**

- §11.4.27 mandates 100% coverage with every test type. This campaign added unit + integration + stress/chaos (partial) + real E2E proofs; the full matrix (ddos/scaling/perf/benchmark/ui/ux, Challenges bank, full HelixQA autonomous sessions) is unfilled. SUBAGENT-TASK: build the §11.4.25 coverage ledger (feature × platform × test-type × status); fill highest-value gaps (perf/benchmark for the translation pipeline, chaos for distributed/storage, Challenges per shipped feature).

P6.2 — docs/qa/ evidence per shipped feature [AUTONOMOUS] · **PARALLEL-SAFE**

- §11.4.83 requires a recorded e2e transcript per shipped feature. SUBAGENT-TASK: per-feature evidence dirs for the campaign's user-visible fixes.
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P7 — Remaining low-yield hunt surface **(completeness, §11.4.118)**

P7.1 — Un-hunted main packages [AUTONOMOUS, low yield] · **PARALLEL-SAFE**

- Not yet hunted (low complexity / low yield): pkg/language, pkg/models, pkg/modelsbridge, pkg/sshworker, pkg/api (handlers), pkg/deployment, pkg/challenge_runner, pkg/logger, pkg/version, pkg/batch (coordination-batch hunted), cmd/* CLI entry points. SUBAGENT-TASK: continue the reproduce-first/mutation-proven hunt one disjoint package per subagent (3-wide; the host rate-limiter kills 4-wide fan-outs — use probe-then-scale).

P7.2 — Audited CLEAN this campaign (no genuine bug; existing guards mutation-verified real, §11.4.6)

- pkg/translator (root), pkg/events, pkg/websocket, pkg/script, pkg/report, pkg/format, pkg/hardware. **Two UNCONFIRMED Windows-only leads** in pkg/hardware (detectGPU false-positive; vm_stat overflow) flagged for a future Windows-capable session — NOT fabricated fixes.
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Confirmed GREEN / DONE this session (for completeness)

- **38 genuine mutation-proven bug fixes** — CONTINUATION rev 52 has the itemized table with commit SHAs (main: distributed×4, ebook/fb2×4, verifier×2, verification×2, storage, security, markdown, coordination, preparation, grpc, progress; submodules: ×20).
 - §11.4.120-reconciled commit_all.sh guard (skip gitlinks + no self-match), paired-verified.
 - All sweeps GREEN at quiescent checkpoints; build+vet clean; all FF-pushed to both upstreams (no force §11.4.113); parent gitlink pointers synced.
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Execution order (recommended)

1. **Autonomous, parallel-safe NOW (3-wide subagent waves):** P4.1 fix-migration into Fixed.md/DB · P4.3 export audit · P4.4 .gitignore hygiene · P6.2 evidence backfill · P7.1 low-yield package hunt.
2. **After operator input:** P1.0 version number · P1.1/P1.2 keys → allowlist audits · P2.x design decisions · P5.2 sibling-clone discard.
3. **Larger autonomous tracks:** P4.2 more CM-gates · P6.1 test-type matrix + Challenges + HelixQA.
4. **Coordinated:** P5.1 security submodule (confirm no collision); helix_qa owned by another session.
5. **Final (release):** P0.2 full §11.4.40 7-step retest → P0.3 §11.4.151 helix_translate-<version> tag fanned to all upstreams.

Nothing here is silently dropped. Every item is tracked to a subagent task with rock-solid-proof acceptance (§11.4.123) and no-bluff verification (§11.4). "Zero issues when we finish" = every P0 closed, every P1/P2 operator-decided, P4–P7 driven to completion or explicitly operator-deferred.